Rahil Parikh

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Robotics engineer working on contact-rich dynamics and optimization-based control, with the systems background to run it on hardware. Built whole-body balance control for a full-size bipedal humanoid, then spent nearly three years shipping production real-time C++ control software — a 47,000-line closed-loop stack with FPGA safety interlocks holding deterministic sub-75 ms guarantees in daily use at customer research facilities. Currently working through contact-implicit trajectory optimization in Drake and training deep RL locomotion policies in NVIDIA Isaac Lab. Dual B.S. Robotics Engineering & Computer Science, WPI (with distinction); incoming M.S.E. Robotics, Penn GRASP, Fall 2026.

Education

University of Pennsylvania, Philadelphia, PA Incoming Fall 2026

M.S.E. in Robotics (GRASP Lab), School of Engineering and Applied Science Expected 2028

- **Planned coursework**: MEAM 5170 Control & Optimization with Applications in Robotics, CIS 5800 Machine Perception, ESE 6500 Learning in Robotics, MEAM 6200 Advanced Robotics

Worcester Polytechnic Institute, Worcester, MA Aug 2019 – May 2023

Dual B.S. in Robotics Engineering and Computer Science — *With Distinction*, GPA: 3.8/4.0

- **Coursework**: Control Engineering, Robot Dynamics, Unified Robotics I–IV, Linear Algebra, Probability, Applied Statistics, Machine Learning, Artificial Intelligence (graduate-level), Embedded Systems, Algorithm Design
- WPI Presidential Scholarship; Dean's List; SWEET Fellowship (research dedication and technical communication); Upsilon Pi Epsilon

Research & Robotics Projects

Contact-Implicit Trajectory Optimization | *Drake, C++, Python* July 2026 – Present

- Working through the C3 (Consensus Complementarity Control) formulation for multi-contact MPC in Drake — complementarity constraints, the ADMM-style consensus split, and the real-time solve budget that makes contact-implicit MPC tractable on hardware
- Motivation: moving from reward shaping toward contact-explicit formulation, after empirically measuring how far a shaped reward can drift from the contact behavior it was meant to produce (see below)

HURON — Full-Size Bipedal Humanoid | *Senior Capstone (MQP) | C++, ROS, Gazebo* Aug 2022 – May 2023

- Led a 12-person interdisciplinary team building a self-balancing bipedal humanoid (lower body) for dynamic walking; secured $15,000 in prototype funding through a faculty advisory pitch cycle
- Owned the controls stack for an underactuated biped: foot-force stability-margin analysis for balance response against external disturbance, and inverse-kinematics gait generation with ZMP-based stability feedback, achieving sub-50 ms reactive balance on hardware and sustained partial walking
- Built the hardware–software interface end to end — actuator calibration, contact and force sensing, and the real-time loop coupling balance feedback to joint commands

Quadruped Locomotion via Deep RL | *Isaac Lab, RSL-RL, PPO, PyTorch* Nov 2025 – Present

- Trained four PPO locomotion policies on AnymalC across a flat/slope/stairs terrain curriculum, verifying genuine trotting gaits in all four (duty cycle 0.33–0.62, diagonal foot correlation −0.56 to −0.90) through gait analysis on held-out terrain rather than reward curves alone
- Designed a leave-one-out reward ablation quantifying the reward–behavior gap: removing `foot_slip` raised total reward +32.6% while halving walking distance, and `terrain_clearance` — a contact-geometry term — proved most valuable at −20.2% task return when removed
- Diagnosed a silent checkpoint-loading failure in rsl-rl v5 where mismatched state-dict keys loaded untrained weights, masking near-zero locomotion behind a standing-bonus reward; resolved across the pipeline via a shared runner utility

Active Perception for Legged Locomotion | *Isaac Lab, PyTorch* Apr 2026 – Present

- Building a policy-controlled pan-tilt depth camera on a quadruped with an information-gain reward grounded in a terrain-predictor's error on future foothold heights — the policy learns where to look in order to improve its own predictions of upcoming contact geometry
- Designed a five-condition ablation and an evaluation suite measuring foothold anticipation distance, saccade phase-locking against the gait cycle, and gaze–terrain mutual information

Professional Experience

Control Systems Software Engineer Sep 2023 – Apr 2026

Bridge12 Technologies, Natick, MA

- Author and owner of a 47,000-line Qt/C++ real-time control system coordinating 8+ heterogeneous instruments through a multi-device finite state machine; deployed in production and operated daily by non-engineers at customer research facilities
- Enforced safety through FPGA hardware interlocks arbitrating against the software control law — hard constraints override nominal control deterministically inside the loop's timing budget, protecting high-power instrumentation from damage
- Delivered hard real-time search-and-track over 100,000+ data points in under 75 ms via spatial hashing and k-d trees, holding deterministic timing under full system load — the same solve-budget discipline optimization-based control demands on hardware
- Cut closed-loop latency 70% (5.0 s $\rightarrow$ 1.5 s) by replacing synchronous polling with a ZeroMQ asynchronous publisher and Kalman-filtered distributed state estimation, raising usable control bandwidth
- Engineered Yocto-based Linux distributions for NVIDIA Jetson Nano and Orin NX, tuning kernel and boot configuration to eliminate latency jitter on embedded control hardware
- Led on-site deployment, debugging, and operator training across 5+ research sites (incl. Columbia University, Havemeyer Hall), converting ambiguous physics requirements into shippable closed-loop systems

Software Engineering Intern May – Oct 2022

Ekotrope, Boston, MA

- Raised MariaDB transaction throughput 20% through query and schema redesign on a physics-based building energy modeling backend; reduced modeling defect rate 18% with JUnit suites validated against analytical baselines

Additional Projects

Autonomous Mobile Robot Navigation | *C++, ROS, LiDAR, AMCL* 2021

- Built a full autonomy stack (SLAM, particle-filter localization, A*/`move_base` planning) with Kalman-filtered LiDAR–odometry fusion; demonstrated collision-free indoor navigation as lead software developer

Robotic Arm Vision & Motion Planning | *C++, MATLAB, OpenCV* 2021

- Built a forward/inverse-kinematics pipeline with an OpenCV pose-estimation front end driving a 6-DOF manipulator to autonomous color-sorted pick-and-place

Technical Skills

Optimization & Contact: trajectory optimization, model predictive control, contact dynamics, complementarity constraints, underactuated systems, ZMP and stability-margin analysis, hybrid dynamics, Drake

Control & Estimation: real-time and safety-critical control, deterministic loop design, feedback control, PID, Kalman filtering, state estimation, inverse kinematics, SO(3)/SE(3) Lie groups

Embedded & Systems: C++ (production, multithreaded), NVIDIA Jetson (Nano, Orin NX), embedded Linux, Yocto, FPGA hardware interlocks, Qt/QML, ZeroMQ, ROS/ROS2, Docker, Git

Robot Learning & Simulation: NVIDIA Isaac Lab, Isaac Gym, MuJoCo, Gazebo, RSL-RL, PPO, reward shaping, terrain curricula, sim-to-real transfer, domain randomization

Perception: PyTorch, OpenCV, depth and point cloud processing, LiDAR sensor fusion, camera calibration, pose estimation

Languages: C++, Python, MATLAB, CUDA, Java, SQL